

Digital Forensics

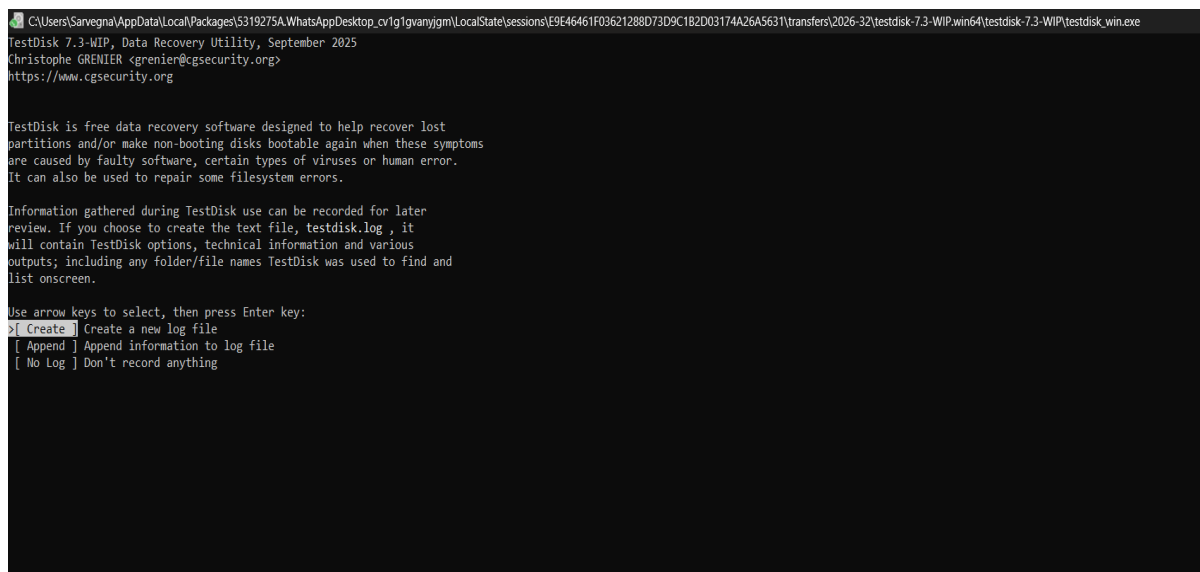
Ex. No: 2 Recover deleted or damaged files from a storage device using Test Disk

Date:

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation



```
C:\Users\Sarvegna\AppData\Local\Packages\S319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\E9E46461F03621288D73D9C182D03174A26A5631\transfers\2026-32\testdisk-7.3-WIP\win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

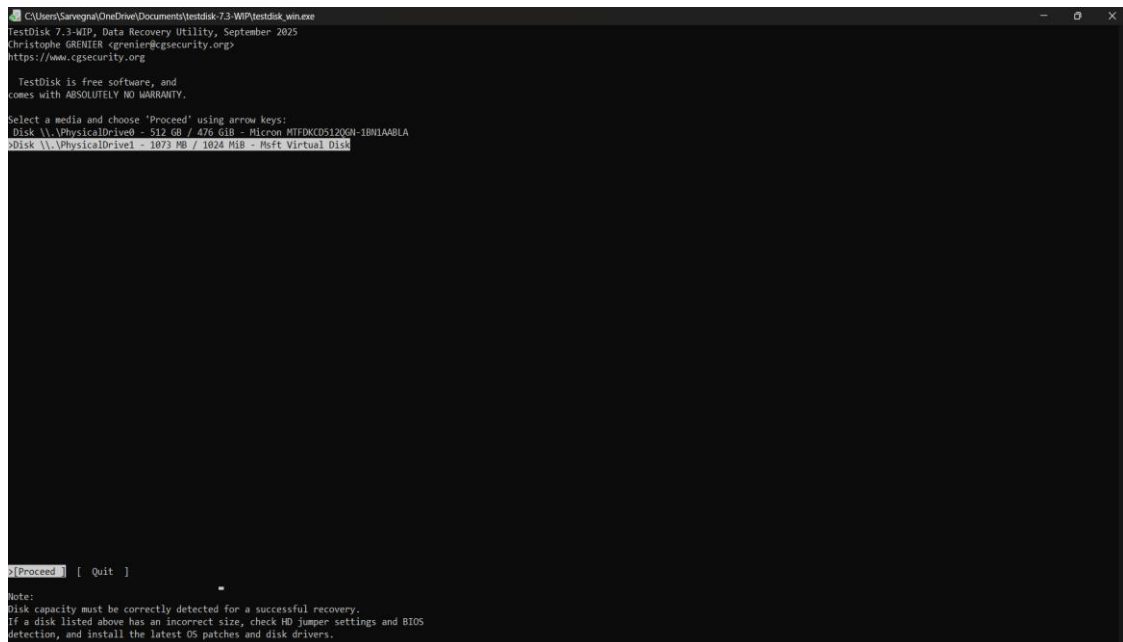
Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything
```

- Choose Create to instruct Test disk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute Test Disk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Test disk was started from a read-only location).
- Press Enter to proceed.

Disk selection

All hard drives should be detected and listed with the correct size by Test Disk:

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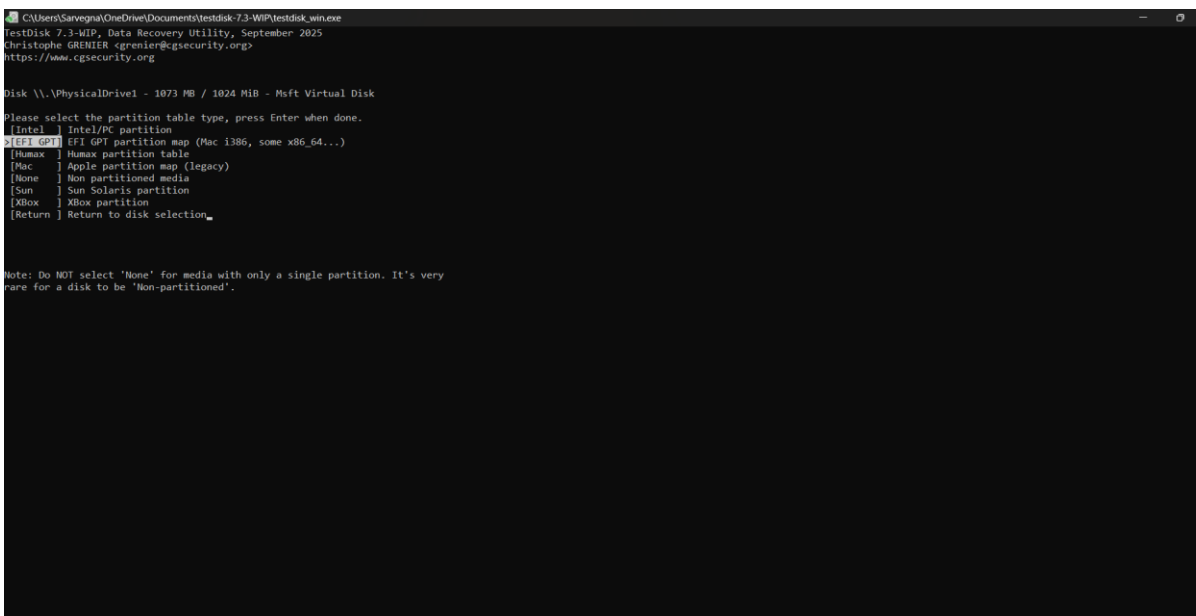


- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

X If available, use raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.



- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

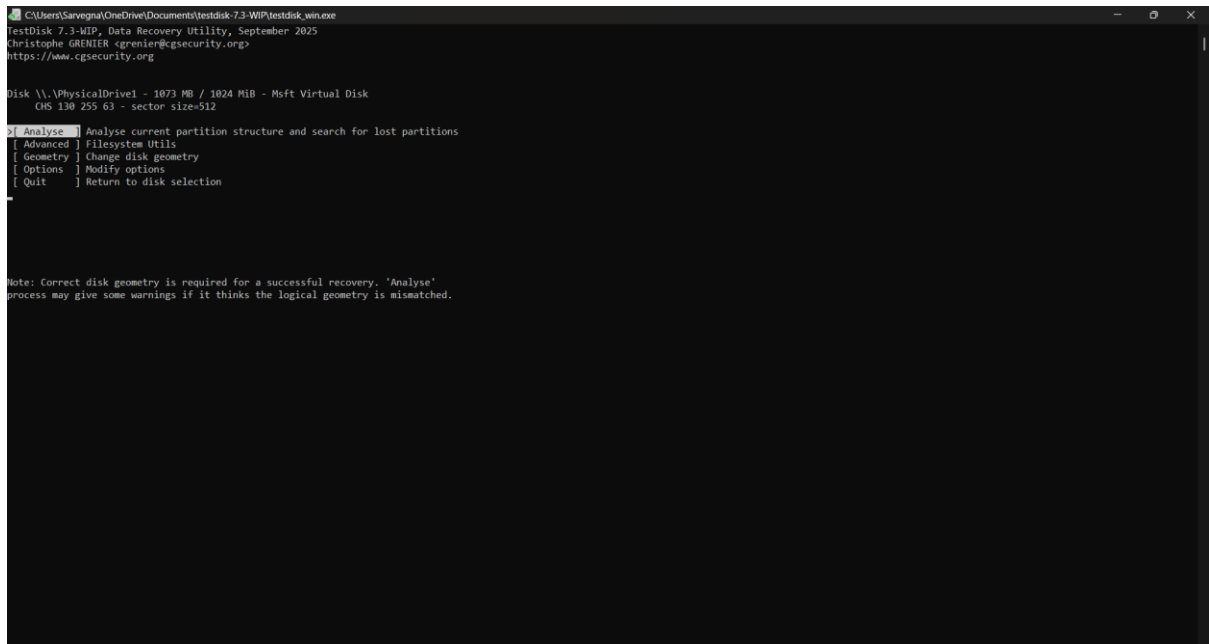
Current partition table status

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TestDisk displays the menus (also see [TestDisk Menu Items](#)).

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.



The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

TestDisk displays the first results in real time. (click on thumb to display the image).

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled **Partition 3**.

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```
CAUsers\Sarvegna\OneDrive\Documents\Testdisk-7.3-WIP\testdisk_win.exe
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Disk \\.\PhysicalDrive1 - 1073 MB / 1024 MiB - CHS 130 255 63
Current partition structure:
    Partition      Start      End      Size in sectors
Bad GPT partition, invalid signature.
Trying alternate GPT
Bad GPT partition, invalid signature.

P=Primary D=Deleted
[Quick Search] Try to locate partition
```

```
CAUsers\Sarvegna\OneDrive\Documents\Testdisk-7.3-WIP\testdisk_win.exe
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Disk \\.\PhysicalDrive1 - 1073 MB / 1024 MiB - CHS 130 255 63
Current partition structure:
    Partition      Start      End      Size in sectors
No FAT, NTFS, ext2, JFS, Reiser, cramfs or XFS marker
1 P MS Reserved    34      32767    32734 [Microsoft reserved partition]
1 P MS Reserved    34      32767    32734 [Microsoft reserved partition]

P=Primary D=Deleted
[Quick Search] [ Backup ] Try to locate partition
```

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

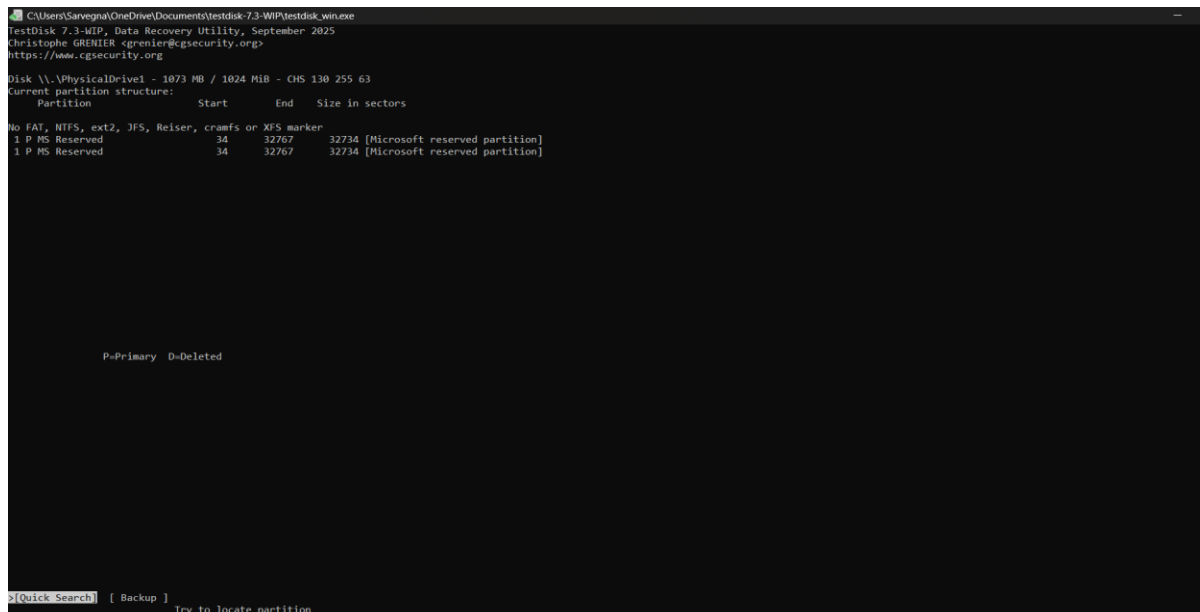
All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

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Y



- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu **Extd Part** gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

A partition is still missing: Deeper Search

Deeper Search will also search for FAT32 backup boot sector, NTFS backup boot superblock, ext2/ext3 backup superblock to detect more partitions,

it will scan each cylinder  (click on thumb).

After the Deeper Search, the results are displayed as follows:

The first partition "**Partition 1**" was found by using backup boot sector. In the last line of your display, you can read the message "**NTFS found using backup sector!**" and the size of your partition. The "partition 2" is displayed twice with different size.

Partitions listed as (Deleted) will not be recovered if you let them listed as deleted. Both partitions are listed with status **D** for deleted, because they overlap each other. You need to identify which partition to recover.

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```
C:\Users\Sarvega\OneDrive\Documents\testdisk-7.3-WIP\testdisk_win.exe
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https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 1073 MB / 1024 MiB - CHS 130 255 63
Partition      Start      End      Size in sectors
+-----+-----+-----+-----+
P: MS Data      32768     2093055   2060288 [New Volume]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
P=Primary D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 1054 MB / 1006 MiB
```

```
C:\Users\Sarvega\OneDrive\Documents\testdisk-7.3-WIP\testdisk_win.exe
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P: MS Data      32768     2093055   2060288 [New Volume]
Directory /

dr-xr-xr-x  0  0      0 10-Aug-2026 17:02 .
dr-xr-xr-x  0  0      0 10-Aug-2026 17:02 ..
dr-xr-xr-x  0  0      0 10-Aug-2026 17:02 $RECYCLE.BIN
dr-xr-xr-x  0  0      0 10-Aug-2026 17:03 Evidence
dr-xr-xr-x  0  0      0 10-Aug-2026 17:00 System Volume Information

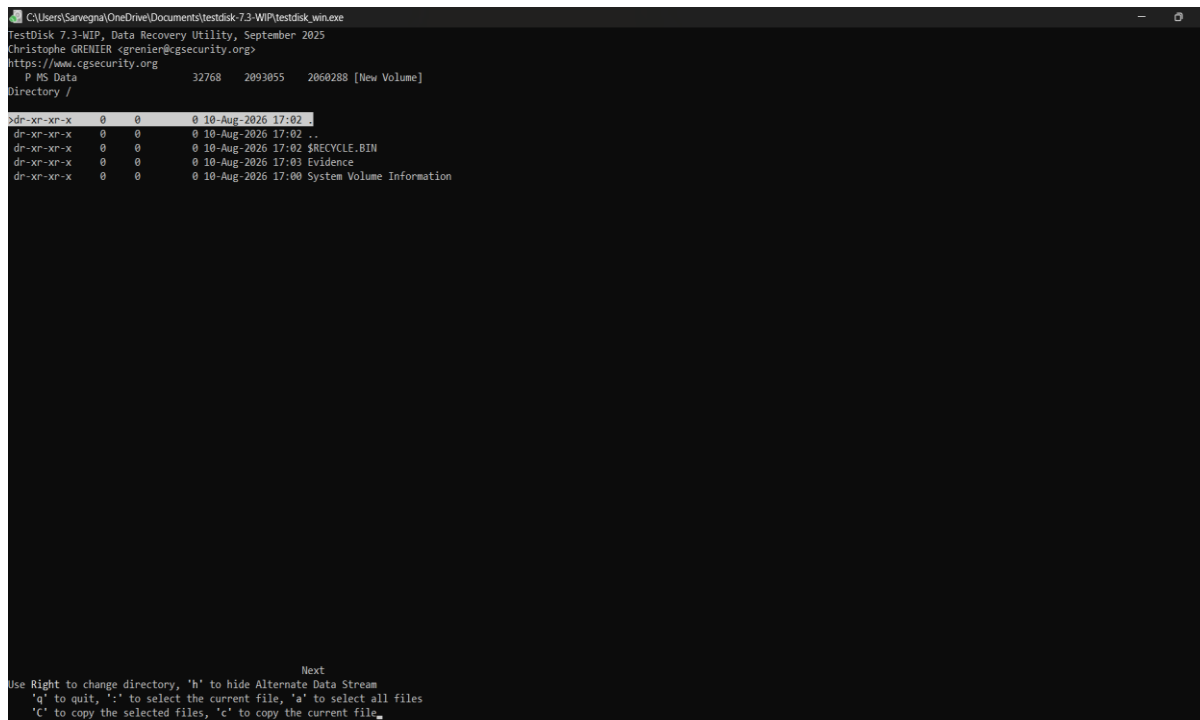
Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'c' to copy the selected files, 'c' to copy the current file
```

- Highlight the first partition **Partition 2** and press **p** to list its data.

The file system of the upper logical partition (label Partition 2) is  (click on thumb).

- Press q for Quit to go back to the previous display.
- Let this partition **Partition 2** with a damaged file system marked as **D(deleted)**.
- Highlight the second partition **Partition 2** below
- Press p to list its files.

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```
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https://www.cgsecurity.org
P MS Data 32768 2093055 2060288 [New Volume]
Directory /

dr-xr-xr-x 0 0 0 10-Aug-2026 17:02 .
dr-xr-xr-x 0 0 0 10-Aug-2026 17:02 ..
dr-xr-xr-x 0 0 0 10-Aug-2026 17:02 $RECYCLE.BIN
dr-xr-xr-x 0 0 0 10-Aug-2026 17:03 Evidence
dr-xr-xr-x 0 0 0 10-Aug-2026 17:00 System Volume Information

Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from (Deleted) to **(Logical)**. This way you will be able to recover this partition.

Hint: read [How to recognize primary and logical partitions?](#)

Note: If a partition is listed *(bootable) but if you don't boot from this partition, you can change it to **Primary** partition.

- Press Enter to proceed.

Partition table recovery

It's now possible to write the new partition structure.

Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

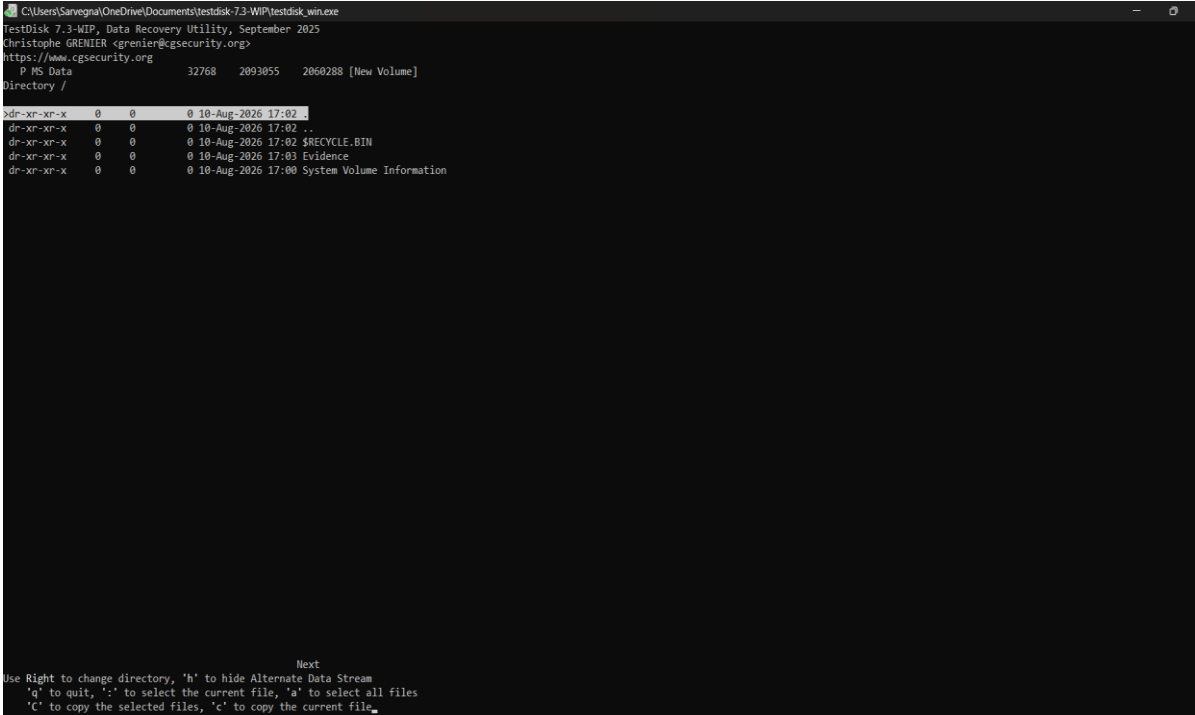
- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

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Now, the partitions are registered in the partition table.

NTFS Boot sector recovery

The boot sector of the first partition named **Partition 1** is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.



```
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TestDisk 7.3-WIP, Data Recovery Utility, September 2025
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https://www.cgsecurity.org
P MS Data 32768 2093055 2060288 [New Volume]
Directory /

sdr-xr-xr-x 0 0 0 10-Aug-2026 17:02
dr-xr-xr-x 0 0 0 10-Aug-2026 17:02 ..
dr-xr-xr-x 0 0 0 10-Aug-2026 17:02 $RECYCLE.BIN
dr-xr-xr-x 0 0 0 10-Aug-2026 17:03 Evidence
dr-xr-xr-x 0 0 0 10-Aug-2026 17:00 System Volume Information

Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

- To copy the backup of the boot sector over the boot sector, select **Backup BS**, validate with Enter, use **y** to confirm and next OK.

More information about repairing your boot sector under [TestDisk Menu Items](#). The following message is displayed:

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```
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Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 1073 MB / 1024 MiB - CHS 130 255 63
Partition      Start          End      Size in sectors
1 P MS Data    32768      2093055  2060288 [New Volume]

Boot sector
Status: OK

Backup boot sector
Status: OK

Sectors are identical.

A valid NTFS Boot sector must be present in order to access
any data; even if the partition is not bootable.

> [Quit] [List] [Rebuild BS] [Repair MFT] [Dump]
Return to Advanced menu
```

The boot sector and its backup are now both OK and identical: the NTFS boot sector has been successfully recovered.

- Press Enter to quit.
- TestDisk displays *You have to restart your Computer to access your data* so press Enter a last time and reboot your compute

RESULT:

The above experiment Recover deleted or damaged files from a storage device using Test disk is successfully completed.